LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 4: Repeat Loops: Do It Again!

Module 2 | 45 minutes | Loops (repeat) | Grades 3–5 · ages 8–10

Objective: Spot repeating patterns and use a repeat loop to run blocks many times, including drawing shapes with the pen.

Success criteria

- I can find a repeating pattern in a script.
- I can use a repeat loop with the right number.
- I can draw a shape with the pen and a loop.

Key vocabulary

- **loop**: Blocks that run again and again.
- **repeat**: A loop that runs a set number of times.
- **pattern**: Something that repeats in a predictable way.
- **pen**: Blocks that let a sprite draw as it moves.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human loop: walk a square with “step, turn × 4”.
5 min	**Watch**	Lesson video; pause on “long way vs loop” and count the blocks.
8 min	**Build along**	Pen + square + triangle (Code Lab or Scratch).
15 min	**Perform**	Hallway puzzle, square in the Code Lab, stairs puzzle, shape-maths blanks.
8 min	**Probe**	Predict, debug Leo’s square, triangle challenge, loops-in-life writing.
4 min	**Wrap-up**	Exit ticket; show one flower from a fast finisher.

Unplugged warm-up

Human loop (5 min): one student is the sprite. The class gives the instructions “step, turn” and the student repeats them 4 times to walk a square. Ask: how could we say this more quickly? (“Do step-turn 4 times.”)

Teaching the idea

A loop repeats a set of instructions. “repeat (n)” runs its inside blocks n times, then the script continues; it shortens code and makes it easier to change. Students first identify the repeating pattern, then place exactly one copy of it inside the loop. With the pen, loops draw regular shapes; the exterior turns sum to 360°.

Analogy: A loop is like the chorus of a song: you write it once and sing it several times.

Live demo: Draw a square the long way (8 blocks) on the board, then ask students to find the pattern. Replace it with a repeat (4). Then change ONE number to draw a triangle — show how easy loops are to change.

Questions to ask

Ask	Listen for
What repeats in this script?	The move and turn pair.
How many times?	Count the sides/stairs.
Why didn’t our square close?	The turn isn’t inside the loop, or the angle isn’t 90.

Differentiation

Support
Give a cut-out loop card students can lay over repeated blocks. Start with the hallway puzzle and the square only.

Stretch
Draw a hexagon, an octagon and a 12-petal flower (loop inside a loop). Predict the turn angle before running.

Common misconceptions

Misconception	How to fix it
Put all the repeated blocks inside the loop (every copy)	Put only ONE copy of the pattern inside, then set the repeat number.
The number in repeat is how far it moves	It is how many times the inside blocks run.
Blocks after the loop run each time too	Only blocks INSIDE the loop repeat; blocks after it run once at the end.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

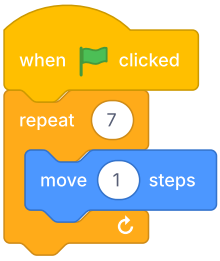
Exit ticket
Write the loop that draws a square. Circle the blocks that are inside the loop.

Homework
Draw (on paper) the shape this makes: repeat 5 { move 100, turn 72 }. Then write the loop for a hexagon.

Answer key — Kitabash activities

1. Long hallway (perform · block code)

One solution:



2. Code Lab: draw a square (perform · code lab)

Goals checked: A closed square · Uses a repeat loop · 8 blocks or fewer

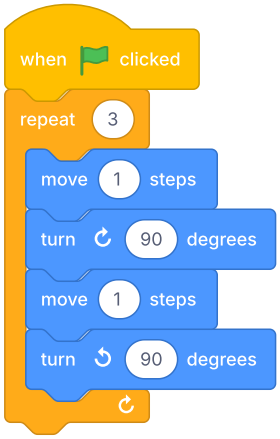
One solution:

Pip



3. Stairs with a loop (perform · block code)

One solution:



4. Shape maths (perform · fill blank)

Answers: 3 · 120 · 6

5. Predict it (probe · multiple choice)

Answer: 3 times

6. Debug it (probe · multiple choice)

Answer: The turn block should be inside the repeat loop

7. Code Lab: triangle (probe · code lab)

Goals checked: A closed triangle · Uses a repeat loop

One solution:

Pip



8. Explain your thinking (probe · free response)

Look for: repeat, times, again. **Example:** Brushing my teeth: I brush each tooth again and again, about 10 times. A song chorus repeats 3 times. A clock ticks forever.

Answer key — worksheet

1. **Shorten it:** repeat (5) { move (10) steps }
2. **Shape maths:** Square 4, 90 · Triangle 3, 120 · Hexagon 6, 60 · Octagon 45
3. **Predict:** Says “Hop!” 4 times; moves 80 steps in total.
4. **Draw it:** A 2 × 1 rectangle: right 2, down 1, left 2, up 1 — back at the start.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.